

Kalman Filter

for Autonomous Car Localization

How a filter blends a prediction with noisy GPS and speedometer readings, explained through four scenarios.

WHAT THIS REPORT COVERS

1

Noisy Prediction

Large process noise makes the prediction wide, so the sensors win.

2

Noisy Measurement

Weak sensors and a tight prediction: the filter barely moves.

3

GPS Outlier Glitch

A 24 m GPS jump meets a confident prediction.

4

Cold Start (Docked)

A stationary car with tiny noise everywhere.

PREDICTION

where the motion model says the car is

MEASUREMENT

what the sensor reports

CORRECTION

the filter's final blended belief

OVERVIEW

How to read the plots

Every curve in this report is a belief about one quantity, written **N(mean, variance)**. Three things are enough to read it:

- **The peak** is the best guess. Where the curve is centered is the estimate.
- **The width** is the uncertainty. A wide, flat curve means “not sure”; a narrow, tall one means “confident”. Every curve holds the same total area, so narrower always means taller.
- **The variance** is the number that sets the width. Its square root is the standard deviation, and about 68% of the belief lies within one standard deviation of the peak.

The filter combines the blue prediction and the red measurement into the green correction. The green curve is always narrower than both, because two sources together leave less uncertainty than either one alone, and its peak leans toward whichever source is more confident.

In one line Position and velocity are linked, because position is built from velocity. Each plot shows only one of them, so a large disagreement in one can nudge the other (see Scenarios 2 and 3).

Experimental setup

Scenario	Start state x_0	Control u	Measurement z	Process noise Q (pos / vel)	Sensor noise R (GPS / speed)
1 Noisy Prediction	[0, 0, 10, 5] $P_0 = 1.0$	[2, 1]	[10.5, 5.2, 9.8, 4.9]	3.0 / 3.0*	2.0 / 2.0*
2 Noisy Measurement	[0, 0, 10, 5] $P_0 = 1.0$	[2, 1]	[10.5, 5.2, 12.0, 6.0]	0.05 / 0.1	9.0 / 9.0
3 GPS Outlier Glitch	[0, 0, 10, 5] $P_0 = 1.0$	[2, 1]	[35.0, 28.0, 10.1, 5.0]	0.05 / 0.1	20.0 / 1.0
4 Cold Start (Docked)	[0, 0, 0, 0] $P_0 = 0.01$	[0, 0]	[0.01, 0, 0, 0]	0.001 / 0.01	0.04 / 0.01

Time step $\Delta t = 1$ s in every scenario. * Scenario 1 uses the corrected values supplied by Aly Sultan.

Results at a glance

Scenario	Plot	Prediction	Measurement	Correction
1 Noisy Prediction	Position X	N(11, 5)	N(10.5, 2)	N(10.54, 1.41)
	Velocity X	N(12, 4)	N(9.8, 2)	N(10.53, 1.32)
2 Noisy Measurement	Position X	N(11, 2.05)	N(10.5, 9)	N(10.91, 1.60)
	Velocity X	N(12, 1.1)	N(12, 9)	N(11.96, 0.91)
3 GPS Outlier Glitch	Position X	N(11, 2.05)	N(35, 20)	N(11.91, 1.46)
	Velocity X	N(12, 1.1)	N(10.1, 1.0)	N(11.55, 0.51)
4 Cold Start (Docked)	Position X	N(0, 0.021)	N(0.01, 0.04)	N(0.003, 0.012)
	Velocity X	N(0, 0.02)	N(0, 0.01)	N(0.000, 0.006)

SCENARIO 1

Noisy Prediction

In one line With the corrected values, the prediction is the wider curve in both plots, so the filter leans on the sensors.

Using the values corrected by Aly Sultan, where the process noise is fixed to 3.0 for both position and velocity and the sensors are set to 2.0, we observe:

Position X

PREDICTION

N(11, 5)

MEASUREMENT

N(10.5, 2)

CORRECTION

N(10.54, 1.41)

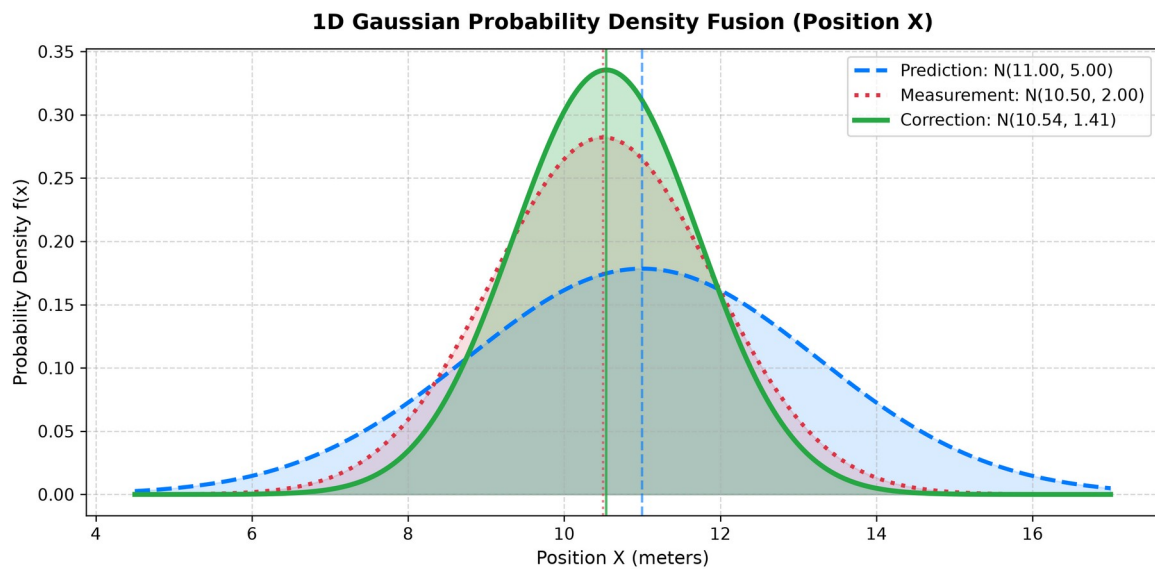


Figure 1a. Scenario 1, Position X

- **Prediction N(11, 5):** The mean is the initial position (0) plus the initial speed multiplied by the time step (10×1), plus the displacement added by the acceleration during that second ($\frac{1}{2} \times 2 \times 1^2$), which gives 11. The variance of 5 ($1 + 1 + 3$) comes mostly from the large process noise, which makes the curve wider and less accurate.
- **Measurement N(10.5, 2):** The mean is the GPS position reading, and the variance of 2.0 is the GPS noise set at the start.
- **Correction N(10.54, 1.41):** The correction sits close to the measurement because it is the more accurate source (variance 2 against 5). It is also narrower than both curves. Blending the two position curves alone would give about 10.64; the rest of the shift comes from the speedometer disagreement.
- **Parameter effects** The process noise of 3.0 widened the prediction from 2 to 5, which makes the measurement the stronger source. With the original values the prediction was the narrower curve, so this scenario only behaves like its name with the corrected values.

Velocity X

PREDICTION
 $N(12, 4)$

MEASUREMENT
 $N(9.8, 2)$

CORRECTION
 $N(10.53, 1.32)$

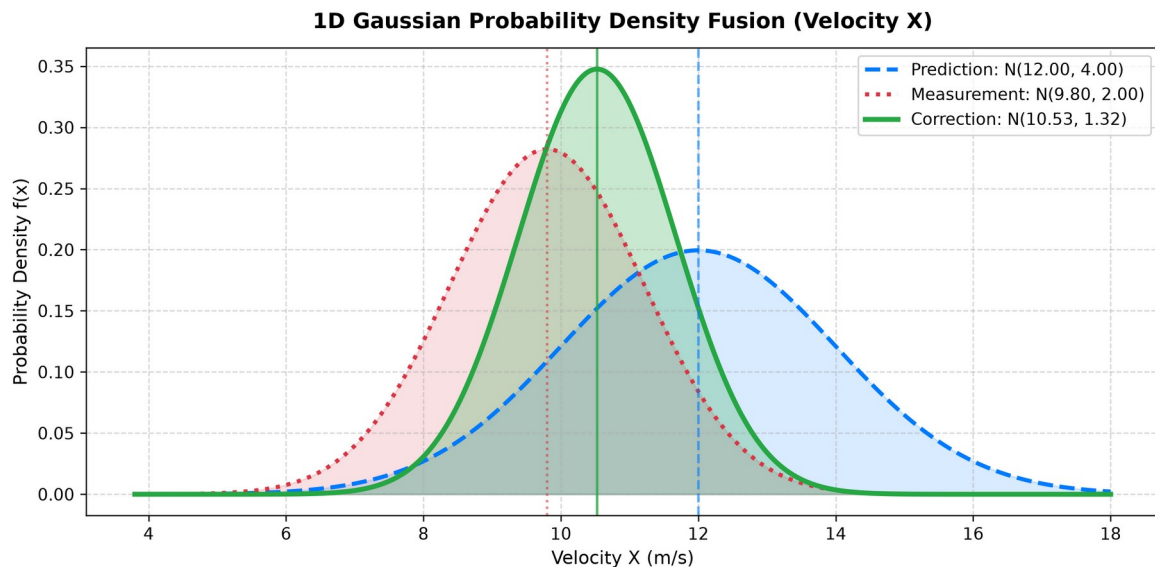


Figure 1b. Scenario 1, Velocity X

- **Prediction $N(12, 4)$:** The mean is the initial speed plus the acceleration over one second, $10 + 2$. The variance is the initial 1.0 plus the process noise of 3.0 , so the curve is wide and the prediction is less reliable.
- **Measurement $N(9.8, 2)$:** The mean is the speedometer reading, and the variance is the speedometer noise $R = 2.0$.
- **Correction $N(10.53, 1.32)$:** The correction sits close to the speedometer reading because it is the more confident source (variance 2 against 4), earning a weight of about 0.66 . This plot behaves like an ordinary blend of the two curves.
- **Parameter effects** The process noise left the prediction with twice the sensor's variance (4 against 2), so the correction is pulled well away from the predicted 12 and toward the speedometer.

SCENARIO 2

Noisy Measurement

In one line The sensors are very noisy ($R = 9$) and the process noise is small, so the filter trusts its prediction and barely moves.

Velocity X

PREDICTION
 $N(12, 1.1)$

MEASUREMENT
 $N(12, 9)$

CORRECTION
 $N(11.96, 0.91)$

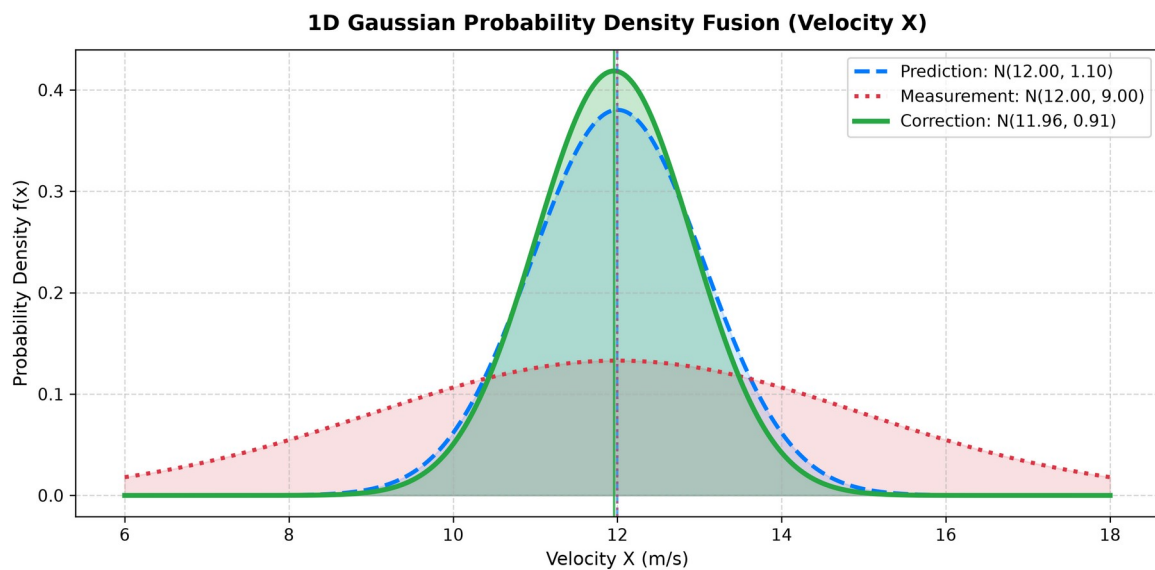


Figure 2a. Scenario 2, Velocity X

The means of the prediction and the measurement align, yet the correction is nudged slightly to the left, to $N(11.96, 0.91)$. The GPS reading differs from the predicted position, and since position and velocity are linked, it suggests the velocity is slightly below 12.

- **Prediction $N(12, 1.1)$:** The mean is $10 + 2$. The variance is the initial 1.0 plus the small process noise of 0.1.
- **Measurement $N(12, 9)$:** The speedometer agrees with the prediction, and its width is set by $R = 9$.
- **Correction $N(11.96, 0.91)$:** With both means at 12 there is nothing to correct from the speedometer, so the small shift to the left comes entirely from the position channel, and the uncertainty shrinks a little.
- **Parameter effects** The speedometer noise of 9 makes it a weak source, so the variance only drops from 1.10 to 0.91. Compared with Scenario 1, the small process noise keeps the prediction tighter.

Position X

PREDICTION
N(11, 2.05)

MEASUREMENT
N(10.5, 9)

CORRECTION
N(10.91, 1.60)

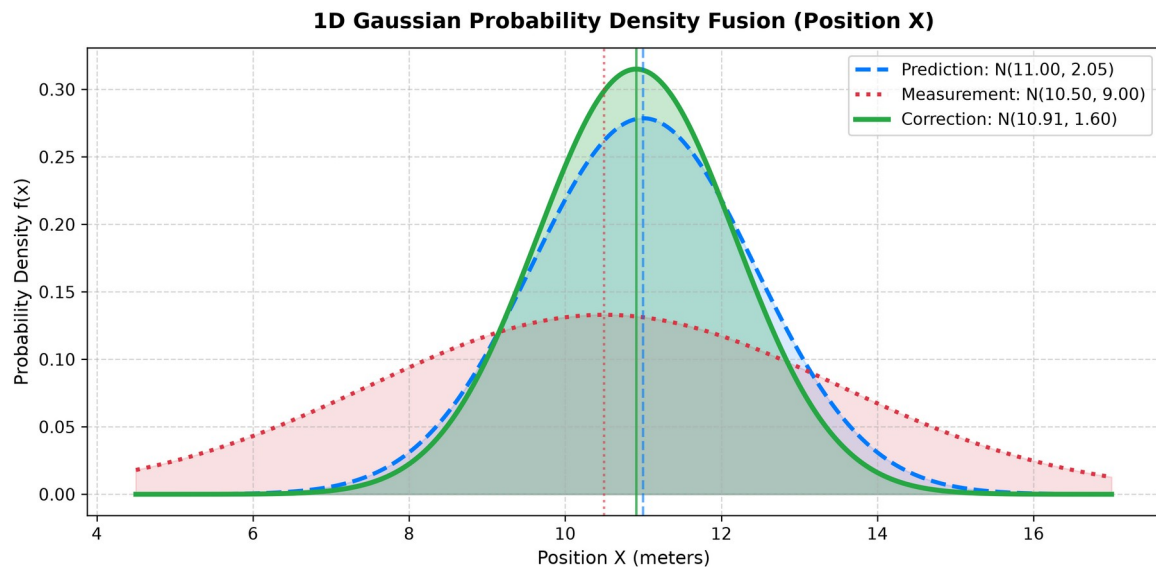


Figure 2b. Scenario 2, Position X

- **Prediction N(11, 2.05):** The mean is $0 + 10 + 1 = 11$. The variance is $1 + 1 + 0.05$, so the prediction is significantly narrower than the measurement.
- **Measurement N(10.5, 9):** The GPS noise R is 9.0, making this a very wide, flat curve.
- **Correction N(10.91, 1.60):** The correction sits close to the prediction mean. The filter moves only about 0.09 of the 0.5 gap between the two.
- **Parameter effects** Small process noise (0.05) and large GPS noise (9) both favor the prediction. The GPS gets a weight of about 0.18, so the correction barely leaves the predicted position.

SCENARIO 3

GPS Outlier Glitch

In one line A GPS reading 24 m away is heavily discounted because its variance is 20, yet it still leaks into the velocity estimate.

Position X

PREDICTION
N(11, 2.05)

MEASUREMENT
N(35, 20)

CORRECTION
N(11.91, 1.46)

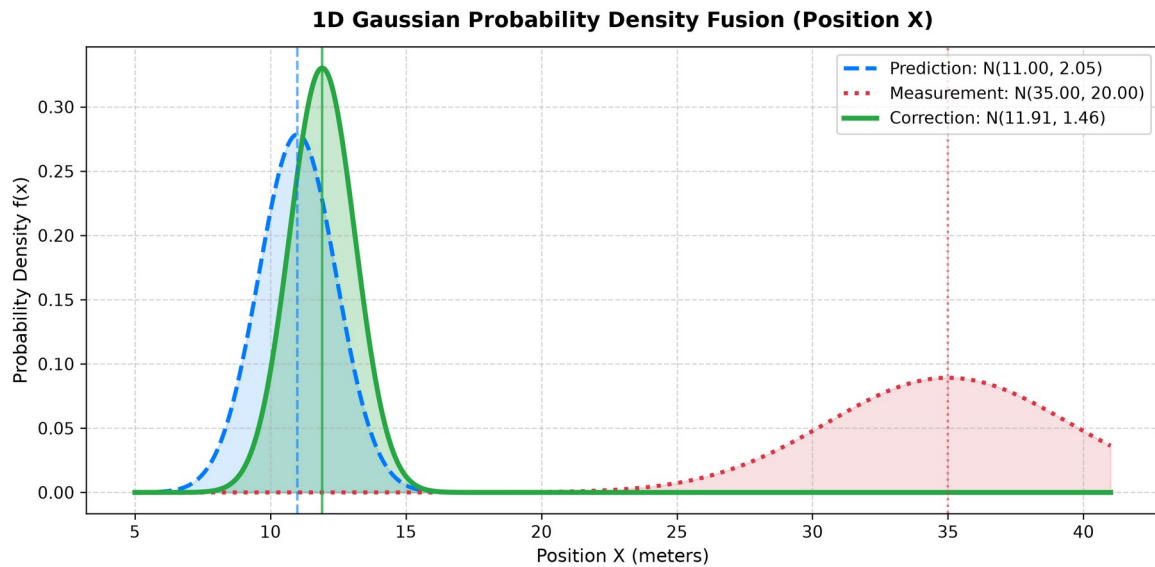


Figure 3a. Scenario 3, Position X

- **Prediction N(11, 2.05):** The mean is $0 + 10 + 1$. The variance is $1 + 1 + 0.05$, the same as in Scenario 2.
- **Measurement N(35, 20):** The GPS reading is 24 m from the prediction, and its very wide curve (variance $R = 20$) says it should not be trusted.
- **Correction N(11.91, 1.46):** The correction stays close to the prediction even though the GPS reads 35. The filter moved less than 1 m toward the glitch, because the GPS noise of 20 marks it as unreliable.
- **Parameter effects** The large GPS noise is what limits the damage. A blend of the two position curves alone would give about 13.2, but the speedometer reads slightly below the predicted speed and pulls position back down.

Velocity X

PREDICTION
 $N(12, 1.1)$

MEASUREMENT
 $N(10.1, 1.0)$

CORRECTION
 $N(11.55, 0.51)$

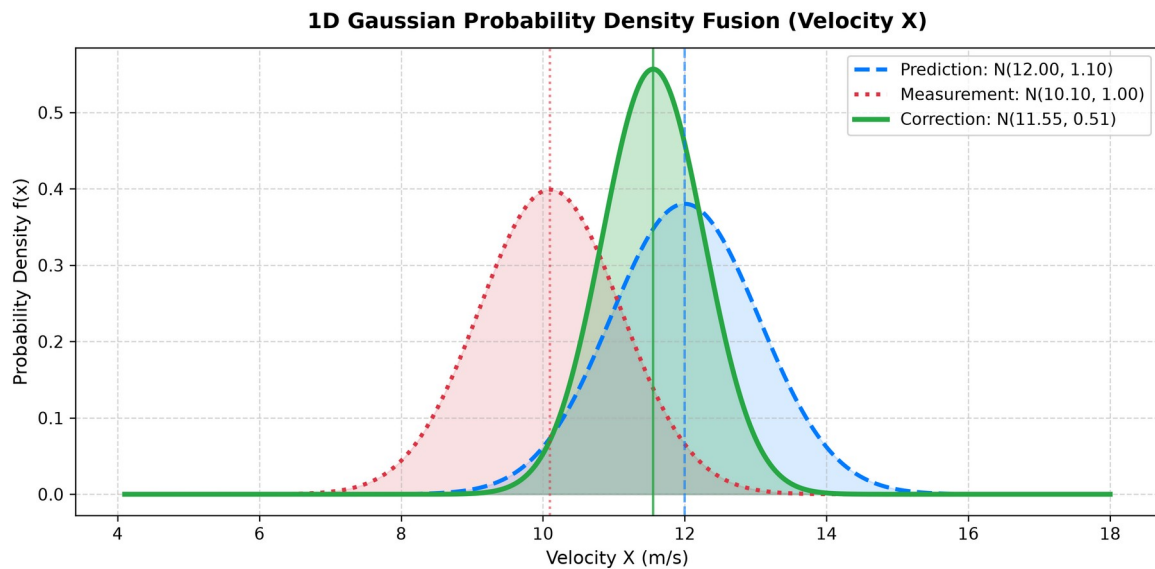


Figure 3b. Scenario 3, Velocity X

- **Prediction $N(12, 1.1)$:** The mean is $10 + 2$. The variance is 1.0 plus the process noise of 0.1 .
- **Measurement $N(10.1, 1.0)$:** The speedometer reading, with variance $R = 1.0$.
- **Correction $N(11.55, 0.51)$:** The correction sits between the prediction and the speedometer. The more precise speedometer alone would pull it to about 11.0 , but the GPS says the car is far ahead, which suggests a higher speed and pushes the estimate back up. So the bad GPS reading still leaks into the velocity.
- **Parameter effects** The small speedometer noise makes it the main source, and the variance drops to 0.51 . A large GPS noise limits the glitch's effect on velocity but cannot remove it.

SCENARIO 4

Cold Start (Docked)

In one line Everything is tiny and the two sources agree, so the mean stays at zero while the uncertainty shrinks.

Position X

PREDICTION
 $N(0, 0.021)$

MEASUREMENT
 $N(0.01, 0.04)$

CORRECTION
 $N(0.003, 0.012)$

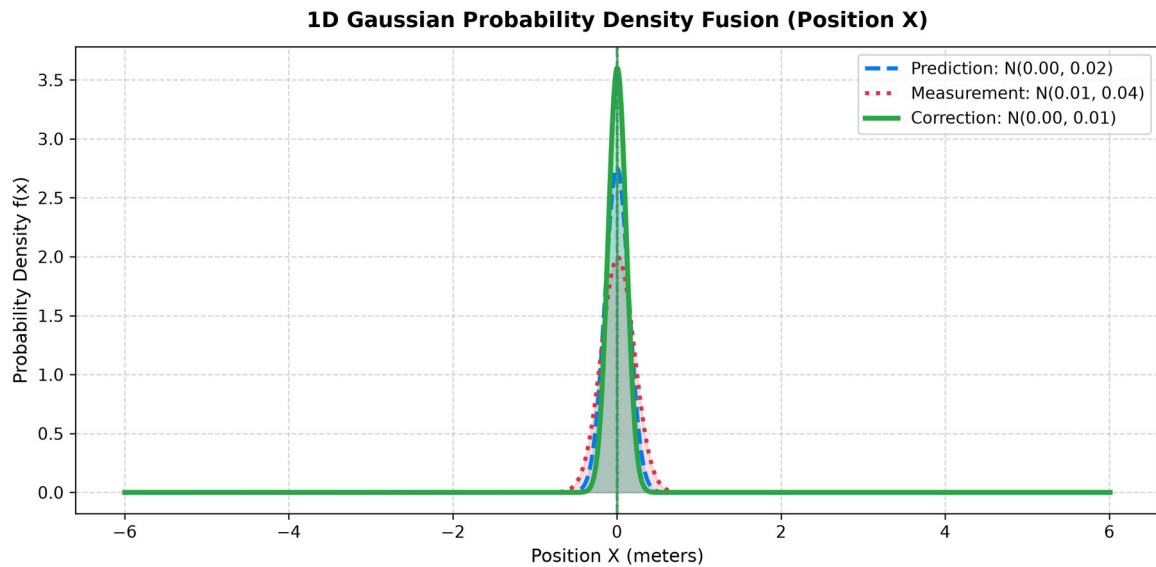


Figure 4a. Scenario 4, Position X

- **Prediction $N(0, 0.021)$:** The car starts at rest with no acceleration, so the mean stays at 0. The variance is $0.01 + 0.01 + 0.001$.
- **Measurement $N(0.01, 0.04)$:** The GPS reading, with variance $R = 0.04$.
- **Correction $N(0.003, 0.012)$:** The correction is essentially at zero. The GPS reading of 0.01 is well within its noise, so the mean barely moves, while the uncertainty shrinks.
- **Parameter effects** Every noise value is tiny, so the curves are narrow and tall. The axes cover a far smaller range than in the other scenarios, which makes the plot look very different.

Velocity X

PREDICTION
 $N(0, 0.02)$

MEASUREMENT
 $N(0, 0.01)$

CORRECTION
 $N(0.000, 0.006)$

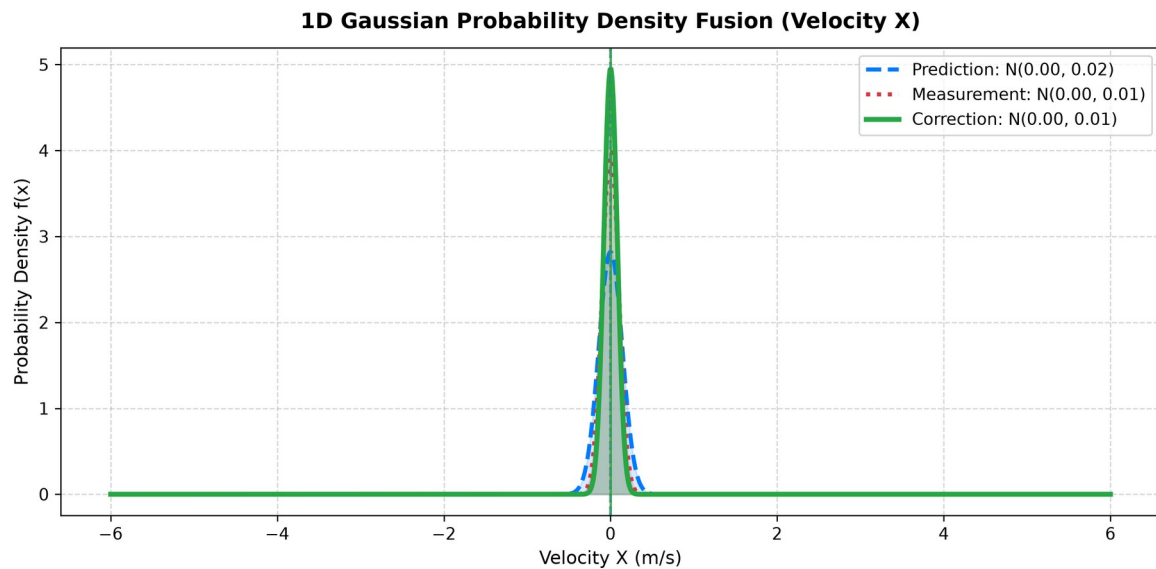


Figure 4b. Scenario 4, Velocity X

- **Prediction $N(0, 0.02)$:** The speed stays 0. The variance is the initial 0.01 plus the process noise of 0.01.
- **Measurement $N(0, 0.01)$:** The speedometer reads 0, with variance $R = 0.01$.
- **Correction $N(0.000, 0.006)$:** This is the tightest curve of the whole set. The two means agree, so nothing moves, and only the variance drops.
- **Parameter effects** The speedometer is more precise than the prediction, so it dominates the combination and gives the largest variance drop of any plot.

Plot legends round to two decimals, so the values above are given to three where the legend would show zeros.

SUMMARY

Key takeaways

- **Always narrower.** The corrected curve is narrower than both the prediction and the measurement in all eight plots. Combining two sources always reduces uncertainty.
- **Leans toward the confident source.** The peak moves toward the narrower curve: toward the sensors in Scenario 1, toward the prediction in Scenario 2.
- **R is a trust setting.** A small sensor noise lets that sensor dominate (the speedometer in Scenarios 3 and 4), while a large one makes it nearly ignored (the GPS in Scenarios 2 and 3).
- **Position and velocity talk to each other.** A disagreement in one quantity can shift the other, as seen in the velocity nudge of Scenario 2 and the leak in Scenario 3.
- **Large noise limits damage; it does not reject.** The GPS glitch in Scenario 3 is heavily discounted but still bends both estimates slightly.